






Time Series Analysis

Chapter 12: Spectral analysis



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **ARMA**: AutoRegressive Moving Average (model)
- **CWT**: Continuous Wavelet Transform
- **DFT**: Discrete Fourier Transform
- **ENTSO-E**: European Network of Transmission System Operators for Electricity
- **EODHD**: EOD Historical Data (market-data provider)
- **EU**: European Union
- **FFT**: Fast Fourier Transform
- **FRED**: Federal Reserve Economic Data
- **GDP**: Gross Domestic Product
- **GPH**: Geweke–Porter-Hudak (estimator)
- **HP**: Hodrick–Prescott (filter)
- **LPPL**: Log-Periodic Power Law
- **MA**: Moving Average
- **OLS**: Ordinary Least Squares
- **SILSO**: Sunspot Index and Long-term Solar Observations
- **STFT**: Short-Time Fourier Transform
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)

The chapter's question and route

- **Question**: which cycles does a series contain, and how much of its variance does each cycle explain?
  - examples: the 11-year sunspot cycle, the business cycle, the daily and weekly rhythm of electricity demand
- **Route** of the chapter
  - cycles, frequencies and the discrete Fourier transform
  - the spectral density of white noise and of ARMA models
  - the periodogram, its weaknesses, and how to smooth it
  - cycles in data; long memory; filters; two series; wavelets as a pointer
- Self-study chapter: worked examples, interpretation slides and a recap after each section; self-assessment with answers at the end

Learning outcomes

- Convert between frequency, period and angular frequency, and find the Fourier frequencies of a sample
- Compute the spectral density of white noise, AR(1), MA(1) and AR(2) models and read its shape
- Compute a periodogram by hand for a short series and explain why the raw periodogram is not consistent
- Smooth the periodogram (Daniell, Bartlett, Welch), taper the data and build chi-square confidence bands
- Identify cycles in sunspots, GDP and electricity load; read coherence and phase between two series

Reading and tools

- Main reference for this chapter: Shumway and Stoffer (2017), Ch. 4 (spectral analysis and filtering)
  - theory: Brockwell and Davis (1991), Ch. 4 and 10; Hamilton (1994), Ch. 6; applications in Python: Huang and Petukhina (2022)
  - Shumway and Stoffer (2017) measure frequency in cycles per observation; Chapter 8 used the angular frequency $\lambda = 2\pi\nu$
- Python Quantlets of this chapter: Quantlets/Ch_12
  - the periodogram, the Daniell smoother and the Morlet wavelet are written out in `numpy`; Welch and coherence come from `scipy.signal`
- Lecture notebook: open in Google Colab
- Video course: Applied Time Series Analysis with Python

Joseph Fourier and the idea of the chapter



J. B. J. Fourier (1768–1830)

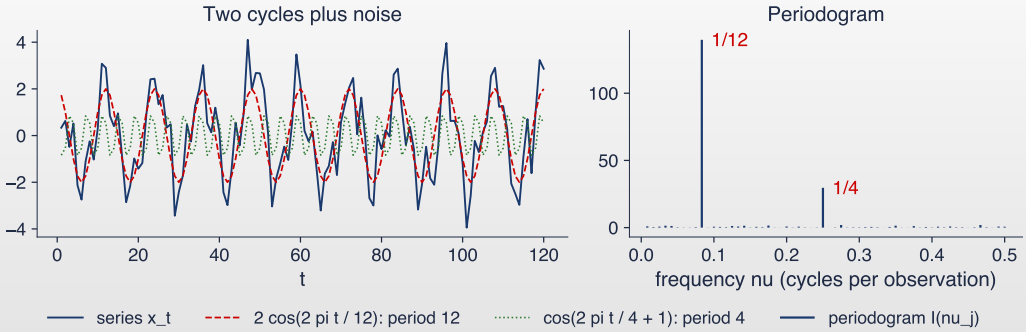
Engraving: A. F. B. Geille, after J.-L. Boilly (1839–1840); public domain; Wikimedia Commons

- Fourier (1807, heat conduction): a function can be written as a sum of sines and cosines
  - each term is a pure cycle with its own frequency and amplitude
- Applied to a time series: how much variance sits at each frequency?
  - the **time domain** (Chapters 1–8) describes the series through its autocorrelations
  - the **frequency domain** describes the same information through the spectrum
- The two views are equivalent; the spectrum makes cycles visible at a glance

A cycle: amplitude, frequency, phase

- A pure cycle: $x_t = A\cos(2\pi\nu t + \varphi)$, $t = 1, 2, \ldots$
  - A: **amplitude** (height of the wave); φ: **phase** (where the wave starts)
  - ν: **frequency**, in cycles per observation; the **period** $1/\nu$ is the number of observations per cycle
  - angular frequency $\omega = 2\pi\nu$, in radians per observation
- Example: monthly data with an annual cycle: $\nu = 1/12$ cycles per month, period 12 months
  - hourly data with a daily cycle: $\nu = 1/24$; quarterly data with a 6-year cycle: $\nu = 1/24$ as well
- Equivalent form: $A\cos(2\pi\nu t + \varphi) = a\cos(2\pi\nu t) + b\sin(2\pi\nu t)$
  - with $a = A\cos\varphi$, $b = -A\sin\varphi$ and $A^2 = a^2 + b^2$: linear in a and b, so it can be fitted by OLS
- The highest frequency visible in discrete data is $\nu = 1/2$ (one cycle every two observations): the **Nyquist frequency**

Two cycles plus noise



- Simulated series, $T = 120$: $x_t = 2\cos(2\pi t/12) + \cos(2\pi t/4 + 1) + \varepsilon_t$, $\varepsilon_t \sim N(0, 0.49)$ (Normal white noise with variance 0.49); right: its periodogram (defined in Section 3)

TSA_ch12_fourier

Interpreting two cycles plus noise

- In the time domain the two cycles are hard to separate by eye
  - in the frequency domain they are two isolated spikes, at $\nu = 1/12$ and $\nu = 1/4$
- Height of the spikes: 139.7 and 29.6; theory for a cycle of amplitude A at a Fourier frequency: $A^2 T/4$, i.e. 120 and 30
  - the spike grows with the squared amplitude: the larger cycle dominates
- The two spikes hold 85% of the total power; the noise spreads the rest evenly over all frequencies

Fourier frequencies and the discrete Fourier transform (1/2)

- For a sample $x_1, \ldots, x_T$, the **Fourier frequencies** are $\nu_j = j/T$, $j = 0, 1, \ldots, \lfloor T/2 \rfloor$
  - cycles that fit an integer number of times (j times) in the sample; $\lfloor T/2 \rfloor$: the integer part of $T/2$
- The **discrete Fourier transform** (DFT) measures how strongly the series moves with a cycle of frequency ν_j
  - $d(\nu_j) = T^{-1/2} \sum_{t=1}^{T} x_t \, e^{-2\pi i \nu_j t}$
  - i: the imaginary unit, $i^2 = -1$; $e^{-i\theta} = \cos\theta - i \sin\theta$ (Euler's formula)
  - so the DFT correlates the series with a cosine (real part) and a sine (imaginary part) of frequency ν_j

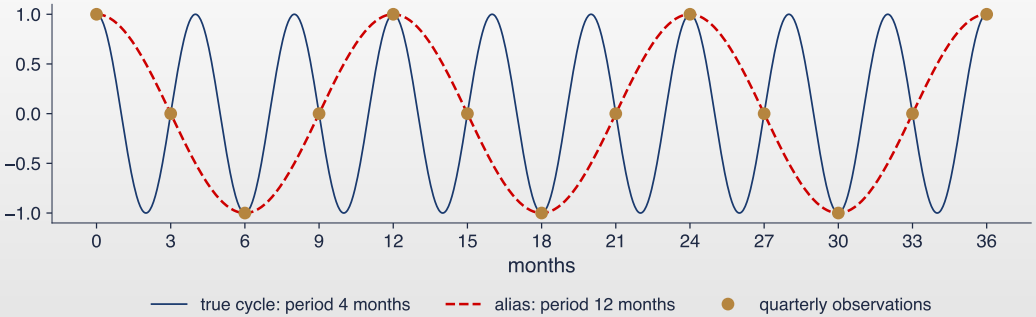
Fourier frequencies and the discrete Fourier transform (2/2)

- The T values $d(\nu_j)$ contain exactly the same information as the T data points
  - the data can be recovered from them by the inverse transform
- **Parseval**
  - the sum of squared deviations equals the sum of the squared moduli of the DFT
  - $\sum_{t=1}^{T}(x_t - \bar{x})^2 = \sum_{j=1}^{T-1} |d(\nu_j)|^2$
  - $\bar{x}$: the sample mean; $|d|^2 = (\mathrm{Re}\, d)^2 + (\mathrm{Im}\, d)^2$: the squared modulus of a complex number
  - reading: the variance of the series is split over the frequencies
- The **fast Fourier transform** (FFT) of Cooley and Tukey (1965) computes all $d(\nu_j)$ quickly
  - $O(T \log T)$ operations instead of $O(T^2)$: the number of operations grows like $T \log T$ instead of T^2

Worked example: the DFT of four numbers

- Data: $x = (4, 2, 0, 2)$, $T = 4$; mean $\bar{x} = 2$, deviations $y = (2, 0, -2, 0)$
  - Fourier frequencies: $\nu_1 = 1/4$ (period 4) and $\nu_2 = 1/2$ (period 2)
- At $\nu = 1/4$: $e^{-2\pi i t/4} = e^{-i\pi t/2}$ equals $-i$ for $t = 1$ and i for $t = 3$
  - $\sum_t y_t e^{-i\pi t/2} = 2(-i) + (-2)(i) = -4i$, so $d(1/4) = -4i/\sqrt{4} = -2i$ and $|d(1/4)|^2 = 4$
- At $\nu = 1/2$: $e^{-i\pi t} = (-1)^t$, so $\sum_t y_t(-1)^t = -2 + 2 = 0$ and $|d(1/2)|^2 = 0$
  - by symmetry $|d(3/4)|^2 = |d(1/4)|^2 = 4$
- Parseval: $4 + 0 + 4 = 8 = 2^2 + 0^2 + (-2)^2 + 0^2$; the whole variance ($8/4 = 2$) sits at period 4
- Check in Python: `abs(np.fft.fft([2, 0, -2, 0]))**2 / 4` gives $(0, 4, 0, 4)$

Aliasing



- A cycle of 4 months ($\nu = 1/4$ per month) observed only once a quarter (every 3 months)

TSA_ch12_fourier

Interpreting aliasing

- Per quarter, the true frequency is $3 \times 1/4 = 3/4$ cycles, above the Nyquist frequency $1/2$
  - it is folded back to $|3/4 - 1| = 1/4$ cycles per quarter: a period of 4 quarters, i.e. 12 months
- The quarterly points lie exactly on a 12-month wave: the data cannot tell the two cycles apart
  - this is **aliasing**: cycles faster than two observations appear as slower, false cycles
- Practical rule: sample at least twice per cycle of interest; average (rather than pick) values when you aggregate to a lower frequency

Recap: Cycles and frequencies

- A cycle has an amplitude, a frequency ν (cycles per observation), a period $1/\nu$ and a phase
- The DFT at the Fourier frequencies j/T rewrites the data as a sum of cycles; Parseval splits the variance by frequency
- Only frequencies up to $1/2$ are visible; faster cycles are aliased

From autocovariances to the spectrum

- For a stationary series with autocovariances $\gamma(h)$, $\sum_h |\gamma(h)| < \infty$, the **spectral density** is
  - $f(\nu) = \sum_{h=-\infty}^{\infty} \gamma(h) e^{-2\pi i \nu h} = \gamma(0) + 2\sum_{h=1}^{\infty} \gamma(h)\cos(2\pi\nu h)$, $-1/2 \le \nu \le 1/2$
  - $\gamma(h) = \mathrm{Cov}(x_t, x_{t+h})$: the autocovariance at lag h; $\sum_h |\gamma(h)| < \infty$: the autocovariances die out fast enough (short memory)
  - in words: $f(\nu)$ is a weighted sum of the autocovariances, with weights $\cos(2\pi\nu h)$ that oscillate at frequency ν
- The inverse relation: $\gamma(h) = \int_{-1/2}^{1/2} f(\nu) e^{2\pi i \nu h}\, d\nu$
  - the pair $\gamma \leftrightarrow f$ is the **Wiener–Khinchin** theorem: the autocovariances and the spectrum carry the same information
  - for $h = 0$: $\mathrm{Var}(x_t) = \gamma(0) = \int_{-1/2}^{1/2} f(\nu)\, d\nu$: the area under f is the variance
- Properties: $f(\nu) \ge 0$, $f(-\nu) = f(\nu)$; it is enough to draw f on $[0, 1/2]$
  - $f(\nu)\, d\nu$ is the share of the variance due to cycles with frequencies in $[\nu, \nu + d\nu]$

White noise, AR(1) and MA(1)

- **White noise** w_t, variance σ^2
  - $\gamma(h) = 0$ for $h \neq 0$, so $f(\nu) = \sigma^2$ for every ν
  - a flat spectrum: all frequencies contribute equally, like white light
- **AR(1)** $x_t = \phi x_{t-1} + w_t$, $|\phi| < 1$
  - $f(\nu) = \dfrac{\sigma^2}{1 - 2\phi\cos(2\pi\nu) + \phi^2}$
  - ϕ: the autoregressive coefficient; w_t: white noise with variance σ^2
  - $\phi > 0$: power at low frequencies (smooth, persistent series); $\phi < 0$: power at high frequencies (a zig-zag series)
- **MA(1)** $x_t = w_t + \theta w_{t-1}$
  - $f(\nu) = \sigma^2\,[1 + 2\theta\cos(2\pi\nu) + \theta^2]$
  - θ: the moving-average coefficient; $\theta > 0$ puts power at low frequencies, $\theta < 0$ at high frequencies
  - directly from the definition: $\gamma(0) = \sigma^2(1 + \theta^2)$, $\gamma(1) = \sigma^2\theta$, so $f = \gamma(0) + 2\gamma(1)\cos(2\pi\nu)$

The spectrum of an ARMA model

- **ARMA**(p, q)
  - $\phi(L)x_t = \theta(L)w_t$
  - L: the lag operator, $Lx_t = x_{t-1}$
  - $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$: the AR and MA polynomials
- Spectral density: the two polynomials evaluated at $z = e^{-2\pi i\nu}$
  - $f(\nu) = \sigma^2 \dfrac{|\theta(e^{-2\pi i\nu})|^2}{|\phi(e^{-2\pi i\nu})|^2}$
  - the MA part (numerator) adds power where $|\theta|$ is large; the AR part (denominator) adds power where $|\phi|$ is small
- A ratio of two polynomials in $\cos(2\pi\nu)$: smooth and easy to compute (Brockwell and Davis (1991), Sec. 4.4)

Worked example: AR(1) and MA(1) spectra

- AR(1) with $\phi = 0.6$, $\sigma^2 = 1$
  - $\nu = 0$: $\cos 0 = 1$, $f(0) = 1/(1-0.6)^2 = 1/0.16 = 6.25$
  - $\nu = 1/2$: $\cos \pi = -1$, $f(1/2) = 1/(1+0.6)^2 = 1/2.56 = 0.39$
  - ratio 16: slow cycles carry far more variance than fast ones
- MA(1) with $\theta = 0.5$, $\sigma^2 = 1$
  - $f(0) = (1+0.5)^2 = 2.25$ and $f(1/2) = (1-0.5)^2 = 0.25$
- Check of the area for the AR(1): $\int_{-1/2}^{1/2} f = \gamma(0) = 1/(1-\phi^2) = 1/0.64 = 1.5625$
  - the numerical integral of the formula gives the same value

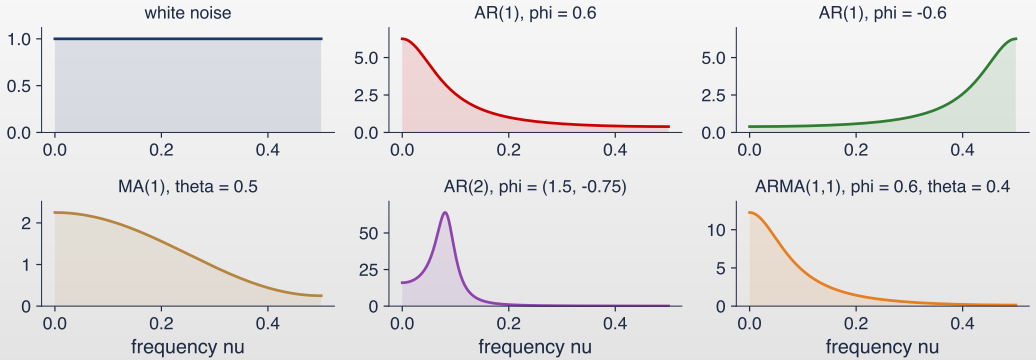
AR(2) and pseudo-cycles

- **AR(2)** $x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + w_t$
  - $f(\nu) = \dfrac{\sigma^2}{|1 - \phi_1 e^{-2\pi i \nu} - \phi_2 e^{-4\pi i \nu}|^2}$
  - with complex roots ($\phi_1^2 + 4\phi_2 < 0$) the ACF is a damped wave and the spectrum has a peak inside $(0, 1/2)$
  - the peak is at the frequency ν^* given by $\cos(2\pi\nu^*) = \phi_1(\phi_2 - 1)/(4\phi_2)$, when the right-hand side lies in $[-1, 1]$
- A **pseudo-cycle**
  - an irregular cycle whose length and amplitude vary from one round to the next
  - Yule (1927) fitted an AR(2) to Wolfer's sunspot numbers: the first autoregressive model, an alternative to fixed sine waves
- Business cycles are pseudo-cycles of this kind: recurrent, but never with the same length (Section 5)

Worked example: the peak of an AR(2) spectrum

- $\phi_1 = 1.5$, $\phi_2 = -0.75$, $\sigma^2 = 1$; complex roots, since $1.5^2 + 4(-0.75) = -0.75 < 0$
  - $\cos(2\pi\nu^*) = 1.5 \cdot (-1.75)/(-3) = 0.875$, so $\nu^* = \arccos(0.875)/(2\pi) = 0.0804$
  - period $1/\nu^* = 12.4$ observations; for yearly data, a cycle of about 12 years
- Height: $f(\nu^*) = 64$ against $f(0) = 1/(1 - 1.5 + 0.75)^2 = 16$
  - a clear peak: the series oscillates with a typical period of about 12 observations
- The same AR(2) is used in Section 4 to test the estimators of the spectrum

A gallery of spectral densities



- Theoretical spectra with $\sigma^2 = 1$ on $0 \leq \nu \leq 1/2$; the shaded area equals half the variance

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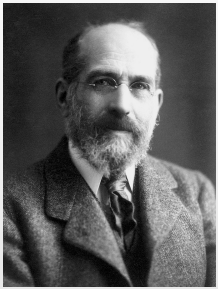
Interpreting the gallery

- Where the power sits tells the type of dependence
  - flat: no dependence (white noise); decreasing: positive persistence (AR(1) with $\phi > 0$, MA(1) with $\theta > 0$, ARMA(1,1))
  - increasing: alternation of signs (AR(1) with $\phi < 0$); a hump inside the band: a pseudo-cycle (AR(2))
- The AR(2) concentrates its variance 8.62 in a narrow band around $\nu = 0.0804$
  - the ARMA(1,1) adds the MA term to the AR(1): $f(0) = 12.25$, a steeper fall
- Reading direction: estimate the spectrum from data, look at its shape, then choose a model whose spectrum looks the same

Recap: The spectral density

- $f(\nu) = \sum_h \gamma(h) e^{-2\pi i \nu h}$; the area under f is the variance; f and the ACF are equivalent
- White noise is flat; AR(1) with $\phi > 0$ puts power at low frequencies; AR(2) with complex roots gives a peak
- ARMA spectra: $\sigma^2 |\theta(e^{-2\pi i \nu})|^2 / |\phi(e^{-2\pi i \nu})|^2$

Arthur Schuster and hidden periodicities



A. Schuster (1851–1934)

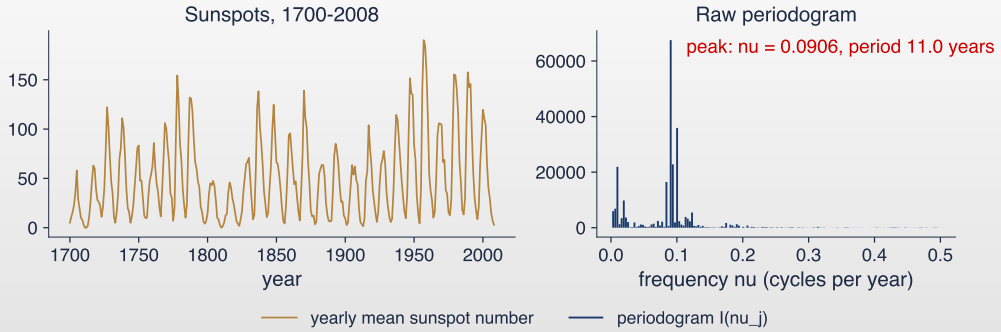
Photo: unknown author (1900s); public domain; Wikimedia Commons

- Schuster (1898) introduced the **periodogram** to search for "hidden periodicities" in meteorological data
  - a claimed 26-day cycle; he showed how to judge whether a peak could be due to chance
- Sunspots: counted daily since the 18th century; Rudolf Wolf built the standard index in 1848
  - their 11-year cycle is the classic test case of spectral analysis (Chapter 0)
- The question of the section: how do we estimate $f(\nu)$ from one sample, and how reliable is the estimate?

The periodogram

- **Periodogram**
  - $I(\nu_j) = |d(\nu_j)|^2$ at the Fourier frequencies $\nu_j = j/T$ (after removing the mean)
  - equivalently $I(\nu_j) = \sum_{|h|<T} \hat{\gamma}(h) e^{-2\pi i \nu_j h}$: the spectral formula with the sample autocovariances $\hat{\gamma}(h)$
- Regression view: regress x_t on $\cos(2\pi\nu_j t)$ and $\sin(2\pi\nu_j t)$ by OLS, with coefficients a_j, b_j
  - $I(\nu_j) = \frac{T}{4}(a_j^2 + b_j^2)$: the squared amplitude of the best-fitting cycle at ν_j, scaled by $T/4$
- Parseval (Section 1) in periodogram form: $\hat{\gamma}(0) = \frac{1}{T} \sum_{j=1}^{T-1} I(\nu_j)$
  - the share of a band of frequencies is the sum of its ordinates divided by the total
- In Python: `np.abs(np.fft.fft(x - x.mean()))**2 / T`; keep $j = 1, \ldots, \lfloor T/2 \rfloor$

Sunspots and their periodogram



⊡ Yearly mean sunspot number, 1700–2008, 309 observations (Wolf / SILSO series, statsmodels); right: the raw periodogram

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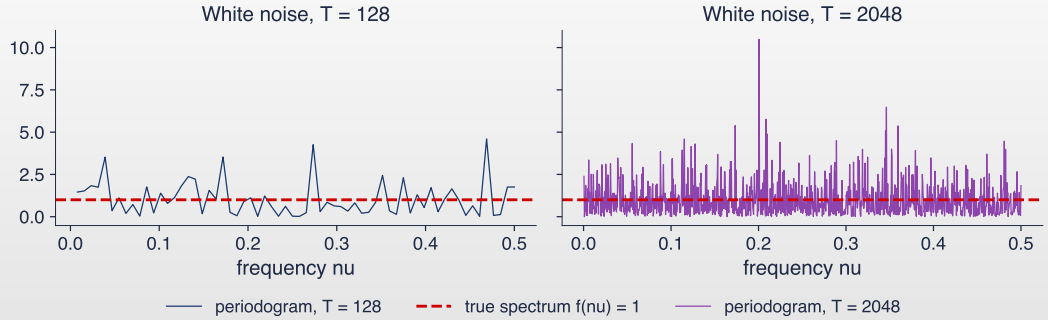
Interpreting the sunspot periodogram

- The highest ordinate is at $j = 28$: $\nu = 28/309 = 0.0906$ cycles per year, a period of 11.0 years
  - the next largest ordinates are at 10.0 and 10.7 years: the cycle is not exactly regular
- Periods between 9 and 13 years carry 60% of the variance
  - the small ordinates near $\nu = 0$ reflect slow changes in the height of the cycles (for example the weak cycles around 1800)
- The raw periodogram is very spiky: neighbouring ordinates jump up and down; the next slides explain why

Statistical properties of the periodogram

- For large T, at a Fourier frequency $0 < \nu_j < 1/2$ (Brockwell and Davis (1991), Sec. 10.3; Shumway and Stoffer (2017), Sec. 4.3):
  - $\frac{2I(\nu_j)}{f(\nu_j)} \approx \chi^2_2$: so $E[I(\nu_j)] \approx f(\nu_j)$ (unbiased) and $\mathrm{Var}[I(\nu_j)] \approx f(\nu_j)^2$
  - χ^2_2: the chi-square distribution with 2 degrees of freedom (mean 2, variance 4); $f(\nu_j)$: the true spectral density
  - ordinates at different Fourier frequencies are approximately independent
- The variance does **not** shrink when T grows
  - the periodogram is **not a consistent** estimator of f
  - a larger T gives more ordinates (a finer grid), not more precise ones
- $\chi^2_2/2$ is an exponential distribution with mean 1: a single ordinate can easily be 3 times the true value
  - its 95% quantile is 3.00: about 1 ordinate in 20 exceeds $3.00\, f(\nu)$ by chance

The periodogram of white noise



- Gaussian white noise with $\sigma^2 = 1$, so the true spectrum is flat at 1; samples of $T = 128$ and $T = 2048$

TSA_ch12_periodogram

Interpreting the white-noise periodogram

- $T = 128$: mean of the ordinates 1.03, standard deviation 1.03
  - $T = 2048$: mean 1.01, standard deviation 1.00: sixteen times more data, the same scatter
- The largest ordinate for $T = 2048$ is 10.5: with 1024 ordinates, isolated tall spikes appear by chance
  - a tall spike is not yet a cycle: it needs a test (next slide) or a smoothed estimate (Section 4)
- The mean is right (about 1): the periodogram is unbiased but noisy; averaging neighbouring ordinates will reduce the noise

Fisher's test of a hidden periodicity (1/2)

- Hypotheses
  - H_0: Gaussian white noise; H_1: white noise plus one cycle at an unknown Fourier frequency
- Statistic of Fisher (1929): the share of the largest ordinate in the sum of all ordinates
  - $g = \max_j I(\nu_j) \,/\, \sum_{j=1}^{m} I(\nu_j)$
  - m: the number of ordinates in $(0, 1/2)$; $\max_j$: the largest of them; $1/m \leq g \leq 1$
- Reading
  - under H_0 each ordinate is about $1/m$ of the total
  - a g much larger than $1/m$ signals a cycle: one frequency takes a large share of the variance

Fisher's test of a hidden periodicity (2/2)

- Exact p-value: the probability, under H_0, of a g above the observed value x
  - $P(g > x) = \sum_{k \geq 1,\ kx<1} (-1)^{k-1} \binom{m}{k} (1 - kx)^{m-1}$
  - k: the summation index, from 1 while $kx < 1$; $\binom{m}{k}$: the binomial coefficient
  - a small p-value (below 0.05): reject white noise in favour of a cycle
- Sunspots: $m = 154$, $g = 0.268$, against $1/m = 0.0065$; p-value < 0.001
  - the 11-year peak is far too large to be chance
- Caution: H_0 is white noise
  - against a red (AR(1)-like) background, low-frequency peaks are more likely by chance
  - so compare with a smooth background spectrum

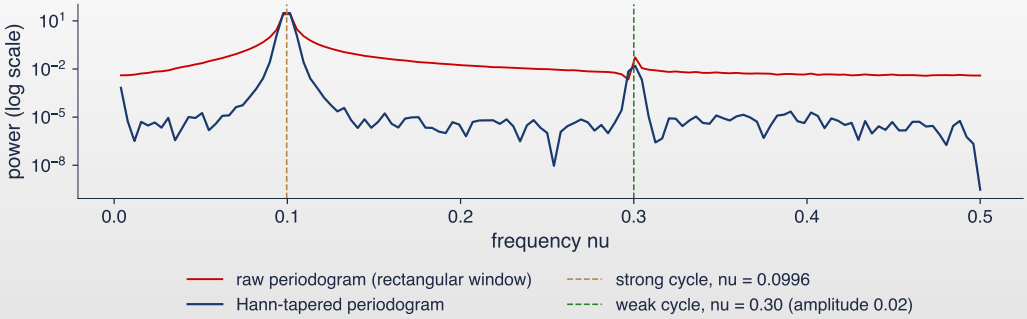
Recap: The periodogram

- $I(\nu_j) = |d(\nu_j)|^2$: proportional to the squared amplitude of the cycle at each Fourier frequency; it sums to the variance
- $2I/f \approx \chi^2_2$: unbiased but not consistent; neighbouring ordinates are nearly independent
- Sunspots: a peak at 11.0 years; Fisher's g rejects white noise

Leakage and tapering

- A cycle whose frequency is not a Fourier frequency does not fit a whole number of times in the sample
  - cutting the series at $t = 1$ and $t = T$ creates a jump; its power **leaks** into all other frequencies
  - leakage from a strong peak can hide a weak cycle elsewhere
- **Tapering**
  - multiply the data by a window h_t that goes smoothly to 0 at both ends before the DFT
  - Hann (cosine bell) window: $h_t = \frac{1}{2}[1 - \cos(2\pi(t - 0.5)/T)]$, rescaled to keep the total power
  - h_t is close to 0 at $t = 1$ and $t = T$ and equals 1 in the middle of the sample
  - the price: a slightly wider main peak (less resolution) for much smaller leakage
- Default in practice: always taper a little; always remove the mean (and a trend, if there is one) first

Leakage: raw and tapered periodograms



- $T = 256$: $\cos(2\pi \cdot 0.0996\, t)$ (between two Fourier frequencies) plus $0.02 \cos(2\pi \cdot 0.3\, t)$ and tiny noise; log scale

TSA_ch12_spectral_estimation

Interpreting leakage

- Raw periodogram: the strong cycle leaks power everywhere; at $\nu = 0.3$ the weak cycle stands only 7.7 times above the leaked background
  - with real noise added, this weak peak would be invisible
- Hann taper: the background falls by several orders of magnitude and the weak cycle stands 4 230 times above it
  - the main peak is a little wider: tapering trades resolution for less leakage
- Log scale is essential: on a linear scale both curves look like a single spike

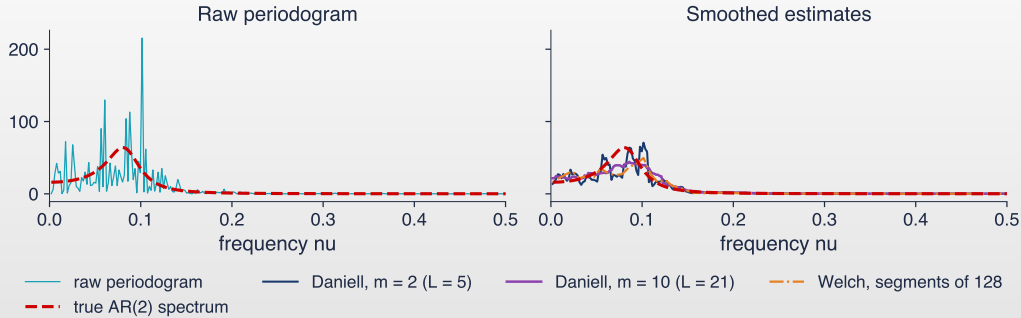
Smoothing the periodogram (1/2): the Daniell estimator

- Idea: f is smooth and the ordinates are nearly independent, so average $L = 2m + 1$ neighbouring ordinates
  - **Daniell** estimator: $\hat{f}(\nu_j) = \frac{1}{L} \sum_{k=-m}^{m} I(\nu_{j+k})$
  - m: the number of neighbours on each side; k: runs from $-m$ to m; $\hat{f}$: the estimated spectrum
  - variance about f^2/L: L times smaller than that of one ordinate
- **Bandwidth** $B = L/T$
  - the width of the frequency band that is averaged
- Bias–variance trade-off: a larger L lowers the variance but flattens narrow peaks (bias)
  - consistency needs $L \to \infty$ and $L/T \to 0$, e.g. $L \approx \sqrt{T}$

Smoothing the periodogram (2/2): lag windows and Welch

- Equivalent view: smoothing in frequency = down-weighting distant autocovariances (lag windows)
  - Bartlett window: $\hat{f}(\nu) = \sum_{|h|\leq M}(1 - |h|/M)\hat{\gamma}(h)e^{-2\pi i\nu h}$ (Bartlett (1950); Blackman and Tukey (1958))
  - M: the truncation lag; autocovariances beyond lag M are dropped, the others get weights $1 - |h|/M$ that fall linearly
  - a smaller M means more smoothing
- The **Welch** method (Welch (1967))
  - an average of the periodograms of K segments: $\hat{f}_W(\nu) = \frac{1}{K}\sum_{k=1}^{K} I_k(\nu)$, I_k: the tapered periodogram of segment k
  - split the series into K overlapping segments, taper each, average their periodograms
  - `scipy.signal.welch`; the segment length sets the resolution

Smoothed estimates of an AR(2) spectrum



- Simulated AR(2), $\phi = (1.5, -0.75)$, $T = 512$; true spectrum (dashed) with the raw periodogram (left) and three smoothed estimates (right)

TSA_ch12_spectral_estimation

Interpreting the smoothed estimates

- True peak at $\nu = 0.080$; the highest raw ordinate is at $\nu = 0.102$: a single noisy ordinate misplaces the peak
  - peak of Daniell $m = 2$: $\nu = 0.102$; of Daniell $m = 10$: $\nu = 0.086$; of Welch ($K = 7$ segments of 128): $\nu = 0.102$
- Mean squared error of $\log \hat{f}$: raw 1.86; Daniell $m = 2$: 0.25; $m = 10$: 0.09; Welch: 0.24
  - the price of heavy smoothing: the peak height drops from 64 to 44 (bias)
- Choose the bandwidth by looking at several values: keep the narrowest one whose estimate is no longer dominated by noise

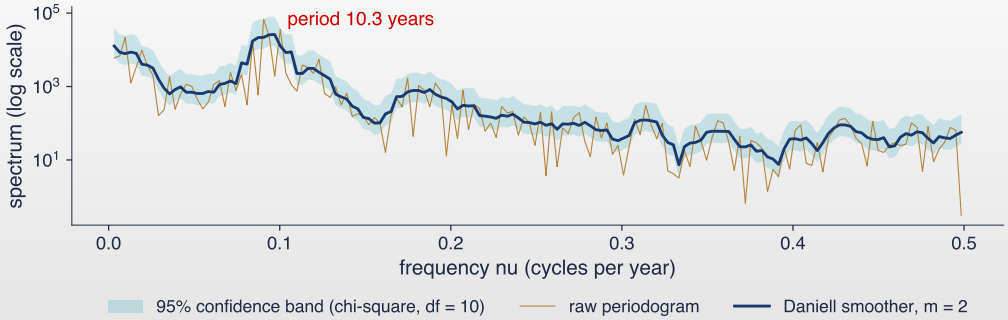
Confidence bands for the spectrum

- Daniell estimate with L terms: $\dfrac{2L\,\hat{f}(\nu)}{f(\nu)} \approx \chi^2_{2L}$ (degrees of freedom $df = 2L$)
  - 95% interval: $\left[\dfrac{df\,\hat{f}(\nu)}{\chi^2_{df}(0.975)},\ \dfrac{df\,\hat{f}(\nu)}{\chi^2_{df}(0.025)}\right]$
  - $\chi^2_{df}(p)$: the quantile of level p of the chi-square distribution with df degrees of freedom
- The interval is a fixed multiple of $\hat{f}$: on a log scale its width is the same at every frequency
  - a peak is significant if the band at the peak lies above the band of the background around it
- Welch with K non-overlapping segments: $df \approx 2K$; overlap and tapering change df a little

Worked example: a 95% band with $L = 5$

- $L = 5$ ($m = 2$), so $df = 10$; from the χ^2_{10} table: $\chi^2_{10}(0.025) = 3.25$ and $\chi^2_{10}(0.975) = 20.48$
  - lower bound: $10/20.48 = 0.49$ times $\hat{f}$; upper bound: $10/3.25 = 3.08$ times $\hat{f}$
- Sunspots: at the smoothed peak (period 10.3 years) $\hat{f} = 26\,095$
  - 95% interval: [12 740, 80 367]: asymmetric, wider upwards
- With only 5 ordinates the estimate is still uncertain by a factor of 2–3; more smoothing narrows the band but blurs the peak

Sunspots: smoothed spectrum with a confidence band



⊡ Daniell smoother with $m = 2$ ($L = 5$, $df = 10$, bandwidth $B = 5/309 = 0.016$ cycles per year) and its 95% band; log scale

TSA_ch12_spectral_estimation

Interpreting the smoothed sunspot spectrum

- The band at the peak (period 10.3 years) lies entirely above the band of the background at higher frequencies
  - the 11-year cycle is a robust feature, not a chance spike
- A second, smaller hump near $\nu \approx 0.18$ (about 5.5 years) is the first **harmonic**
  - the sunspot cycle is not a sine wave: it rises fast and decays slowly, and a non-sinusoidal cycle has power at multiples of its frequency
- Power also rises towards $\nu = 0$: the height of the cycles changes over decades and centuries

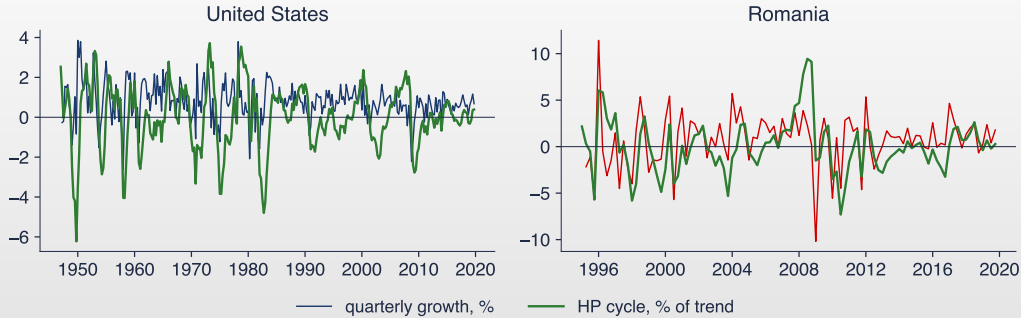
Recap: Estimating the spectrum

- Taper to reduce leakage; plot spectra on a log scale
- Smooth over L ordinates (Daniell, Bartlett, Welch): variance $\approx f^2/L$, but narrow peaks are flattened
- 95% band: $[df\ \hat{f}/\chi^2_{df}(0.975),\ df\ \hat{f}/\chi^2_{df}(0.025)]$ with $df = 2L$

The typical spectral shape and the business cycle

- Granger (1966): the spectrum of most economic levels (GDP, prices) falls steeply from $\nu = 0$: the **typical spectral shape**
  - trends and near unit roots put almost all the variance at the lowest frequencies and hide the cycles
  - so we first make the series stationary: growth rates (differences) or a detrended cycle
- **Business-cycle band**
  - fluctuations lasting 1.5 to 8 years, i.e. 6 to 32 quarters (Baxter and King (1999))
  - in frequency: $1/32 \leq \nu \leq 1/6$ cycles per quarter
- Data: US real GDP (FRED GDPC1, 291 quarterly growth rates) and Romanian real GDP (Eurostat, seasonally adjusted, 99 growth rates)
  - both up to 2019 Q4: the 2020 collapse and rebound would dominate every periodogram
  - the cycle is also extracted with the HP filter, $\lambda = 1600$ (Hodrick and Prescott (1997); Chapter 10)

GDP growth and HP cycles



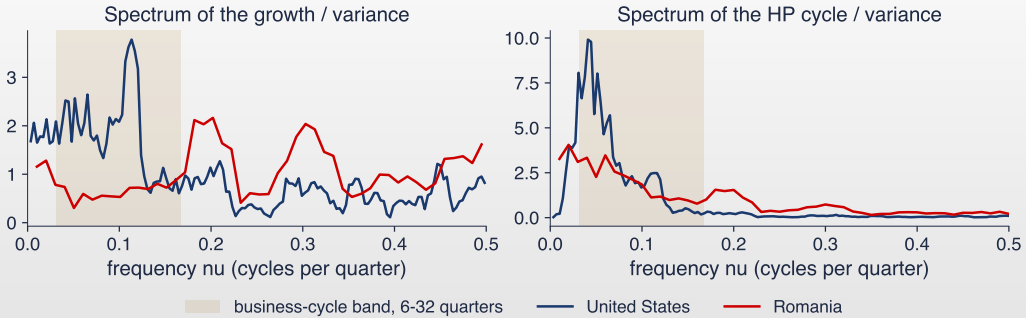
⊡ Quarterly growth $100\,\Delta \ln \mathrm{GDP}_t$ and the HP cycle ($100 \ln \mathrm{GDP}_t$ minus its HP trend), United States 1947–2019 and Romania 1995–2019

TSA_ch12_business_cycle

Interpreting the GDP series

- United States: mean growth 0.78% per quarter, standard deviation 0.93; HP cycle standard deviation 1.58% of trend
  - the cycle shows the recessions as deep troughs (1958, 1975, 1982, 2009)
- Romania: mean growth 0.73%, standard deviation 2.85: three times more volatile
  - one large boom–bust (2004–2010) dominates a sample of only 25 years
- The growth rate is noisy and short-lived; the HP cycle is smooth and persistent: their spectra will differ

Spectra of GDP growth and HP cycles



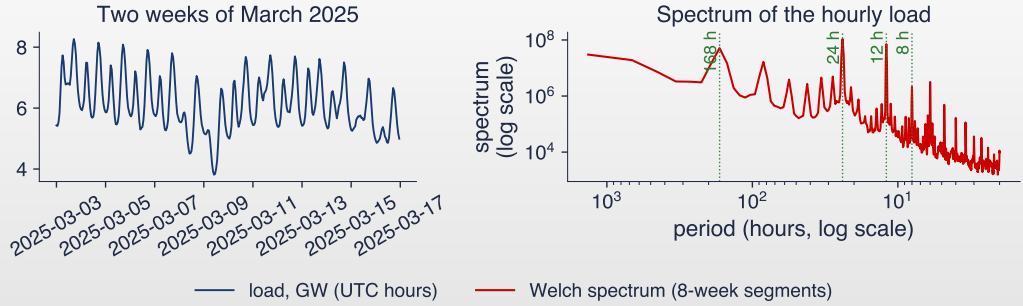
- Daniell smoother, $m = 2$; each spectrum divided by the variance of its series; shaded: periods of 6 to 32 quarters

TSA_ch12_business_cycle

Interpreting the GDP spectra

- US growth: 49% of the variance in the business-cycle band; the smoothed peak is at about 8.8 quarters
  - Romanian growth: only 16%; most of its variance is at high frequencies (quarter-to-quarter noise and revisions)
- US HP cycle: 80% of the variance in the band, peak at 24.3 quarters (about 6.1 years)
  - Romanian HP cycle: 49% in the band and 17% at periods longer than 8 years: one long cycle in a short sample
- Caution: the HP filter itself removes low frequencies and can shape the peak (Section 7); 99 Romanian observations give a very wide confidence band

Spectrum of the Romanian hourly electricity load



- Hourly electricity load of Romania (ENTSO-E, MW, UTC hours), 2022–2026, 39 408 hours, mean 6 176 MW (Chapter 4); Welch spectrum with 8-week segments

TSA_ch12_electricity_load

Interpreting the load spectrum

- Sharp peaks at 24, 12 and 8 hours: the daily cycle and its harmonics (morning and evening peaks are not a sine wave)
  - the daily cycle and its harmonics hold 34% of the variance
- A peak at 167.7 hours (one week) and its harmonic at 84.0 hours: working days against weekends
  - weekly components: 16%; periods longer than a month (winter against summer): 33%
- This is why Chapter 4 modelled the load with daily and weekly Fourier terms: the spectrum shows which periods to include

Recap: Cycles in data

- Remove trends first (growth rates, detrended cycles): levels have the typical spectral shape
- US GDP: a business-cycle peak of about 6.1 years in the HP cycle; Romania: too short a sample for a sharp answer
- Electricity load: daily and weekly cycles with harmonics; the spectrum chooses the Fourier terms of a forecasting model

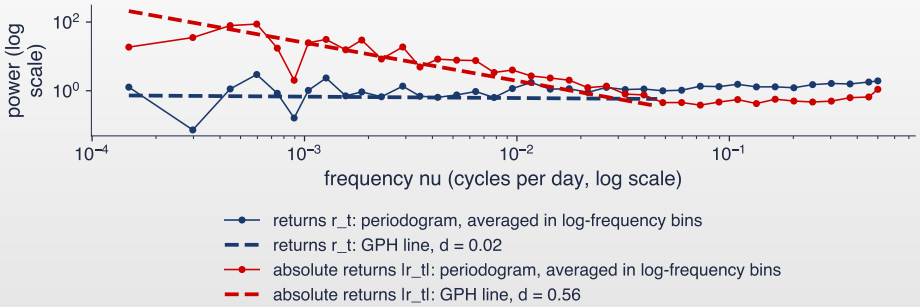
Long memory in the frequency domain (1/2)

- Chapter 8: a long-memory series has autocorrelations that decay like a power of the lag
  - $\rho(h) \sim Ch^{2d-1}$: $\rho(h)$ is the autocorrelation at lag h, $C > 0$ a constant, $\sim$ means "proportional for large h"
  - d: the memory parameter; $0 < d < 1/2$ gives stationary long memory; then $\sum_h \gamma(h) = \infty$ and $f(0) = \infty$
- ARFIMA$(0, d, 0)$: the spectrum explodes at zero
  - $f(\nu) = \sigma^2 \, |2\sin(\pi\nu)|^{-2d} \approx \sigma^2 (2\pi\nu)^{-2d}$ near $\nu = 0$: a **pole** at zero
  - taking logs: $\log f(\nu) \approx \log \sigma^2 - 2d \log(2\pi\nu)$, a straight line in $\log \nu$ with slope $-2d$
- Short memory (ARMA) has a finite $f(0)$: the log–log plot flattens at low frequencies

Long memory in the frequency domain (2/2): the GPH estimator

- Geweke and Porter-Hudak (1983): a regression of the log-periodogram on a log-frequency term
  - regress $\log I(\nu_j)$ on $-\log(4\sin^2(\pi\nu_j))$ for $j = 1, \ldots, m$; the slope is $\hat{d}$
  - standard error $\pi/\sqrt{24m}$: it shrinks as more frequencies are used
- Only the lowest $m = \lfloor T^{0.65} \rfloor$ frequencies are used, where the pole dominates
  - a larger m lowers the standard error but brings in short-memory effects (bias)
- Reading
  - $\hat{d}$ close to 0: short memory; $0 < \hat{d} < 0.5$: stationary long memory; $\hat{d} \geq 0.5$: nonstationary

The pole at zero in stock-market data



- Daily S&P 500 log returns, 2000–2026, $T = 6\,714$ (EODHD); periodogram averaged in 40 log-frequency bins; GPH lines on the lowest $m = 307$ frequencies (periods of 21.9 days and longer)

TSA_ch12_long_memory

Interpreting the log–log periodogram

- Returns r_t: flat at low frequencies, GPH $\hat{d} = 0.02$ (standard error 0.037): no long memory in the returns
  - a flat spectrum is what market efficiency predicts for returns (Chapter 1)
- Absolute returns $|r_t|$: power rises steadily towards $\nu = 0$, slope $-2\hat{d}$ with $\hat{d} = 0.56$
  - the long memory of volatility (Chapter 8); $\hat{d}$ near or above 0.5 also reflects breaks such as 2008 and 2020
- The same spectral picture explains why GPH and local Whittle (Chapter 8) are frequency-domain estimators

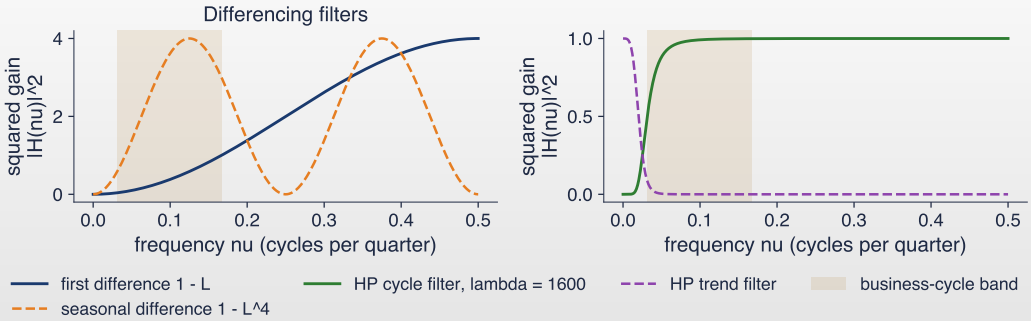
Linear filters and their gain (1/2)

- A **linear filter** turns an input series x_t into an output y_t by a weighted sum of current and past values
  - $y_t = \sum_j a_j x_{t-j}$; a_j: the filter weights; j: the lag
- Its **frequency response** is $A(\nu) = \sum_j a_j e^{-2\pi i \nu j}$
  - key result: $f_y(\nu) = |A(\nu)|^2 f_x(\nu)$; f_x, f_y: the spectra of input and output
  - $|A(\nu)|^2$ is the **squared gain**: above 1 the filter amplifies cycles of frequency ν, below 1 it damps them, 0 removes them

Linear filters and their gain (2/2)

- First difference $y_t = x_t - x_{t-1}$: $|A(\nu)|^2 = |1 - e^{-2\pi i\nu}|^2 = 4\sin^2(\pi\nu)$
  - zero at $\nu = 0$ (removes the trend), 4 at $\nu = 1/2$ (amplifies noise); at a period of 40 quarters: 0.025; at 4 quarters: 2
- Seasonal difference $(1 - L^4)x_t = x_t - x_{t-4}$ (quarterly data): squared gain $4\sin^2(4\pi\nu)$
  - zero at $\nu = 0, 1/4, 1/2$: it removes the trend and the seasonal cycles
- HP cycle filter: gain $G(\nu) = \dfrac{4\lambda(1 - \cos 2\pi\nu)^2}{1 + 4\lambda(1 - \cos 2\pi\nu)^2}$ (real), squared gain $G(\nu)^2$
  - λ: the smoothing parameter (1600 for quarterly data); a larger λ moves the cut-off to slower cycles
  - close to 0 at low frequencies, close to 1 at high ones: a **high-pass** filter

Squared gains of common filters



⊡ Left: first difference and seasonal difference ($s = 4$); right: HP cycle and trend filters, $\lambda = 1600$ (quarterly data); shaded: 6–32 quarters

TSA_ch12_filters

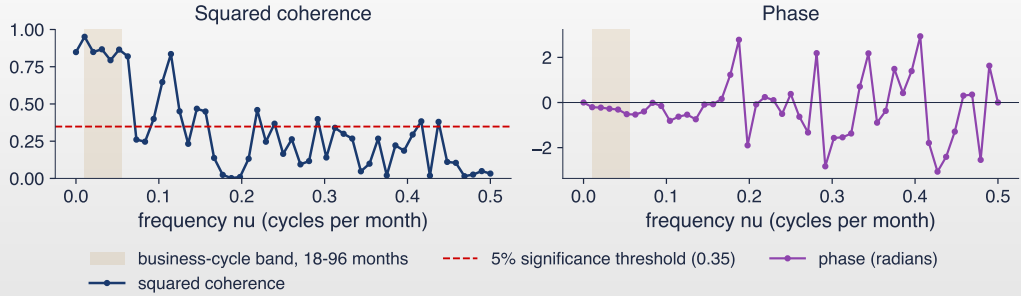
Interpreting the filter gains

- Differencing is not neutral: it weakens business cycles and amplifies the highest frequencies
  - this is why growth rates look noisier than levels; filtering white noise can even create spurious cycles (the Slutsky effect, Chapter 0)
- HP cycle filter: squared gain 0.996 at 8 quarters, 0.49 at 32 quarters, 0.24 at 40 quarters; half power at about 32 quarters
  - it keeps most of the power of cycles faster than about 8 years, including noise: not a band-pass filter (Baxter and King (1999))
- Applied to a random walk, the HP filter produces a cycle with a peak that comes from the filter, not from the data (Hamilton (2018))

Cross-spectrum, coherence and phase

- For two stationary series with cross-covariances $\gamma_{xy}(h) = \mathrm{Cov}(x_{t+h}, y_t)$:
  - **cross-spectrum** $f_{xy}(\nu) = \sum_h \gamma_{xy}(h) e^{-2\pi i \nu h}$, a complex number at each frequency
  - f_x, f_y: the spectra of x and y separately
- **Squared coherence**
  - $\rho^2_{xy}(\nu) = \dfrac{|f_{xy}(\nu)|^2}{f_x(\nu) f_y(\nu)} \in [0, 1]$
  - a squared correlation frequency by frequency: how well the cycles of x and y at ν move together
- **Phase** $\varphi(\nu) = \arg f_{xy}(\nu)$
  - the shift between the two cycles, in radians
  - arg: the angle of a complex number in the plane (its argument)
  - a time lag of k periods gives a phase that is linear in frequency: $\varphi(\nu) = \pm 2\pi\nu k$; so the lag is $|\varphi|/(2\pi\nu)$
- Estimation: smooth the cross-periodogram like the periodogram; the raw coherence is always 1, so smoothing is compulsory

Industrial production and unemployment



- US monthly data, 1948–2019 (FRED INDPRO, UNRATE), $T = 863$: $x_t = 100\,\Delta \ln \mathrm{IP}_t$ and $y_t = -\Delta u_t$ (the fall in the unemployment rate); Welch with $K = 8$ segments of 96 months

TSA_ch12_coherence

Interpreting coherence and phase

- In the business-cycle band (18–96 months) the squared coherence averages 0.87 (maximum 0.95), far above the 5% threshold 0.35
  - the 5% threshold for a Welch estimate with K segments: $1 - 0.05^{1/(K-1)}$
  - at frequencies faster than 6 months it averages only 0.17: the monthly noise of the two series is unrelated
  - the ordinary correlation, 0.49, mixes the strong link of the cycles with the weak link of the noise
- Phase at the 48-month cycle: -0.23 radians, a lag of about 1.7 months
  - the fall in unemployment follows the recovery of production with a short delay (Okun's law, with a lag)
- Where the coherence is low, the phase is meaningless: read the phase only in bands with significant coherence

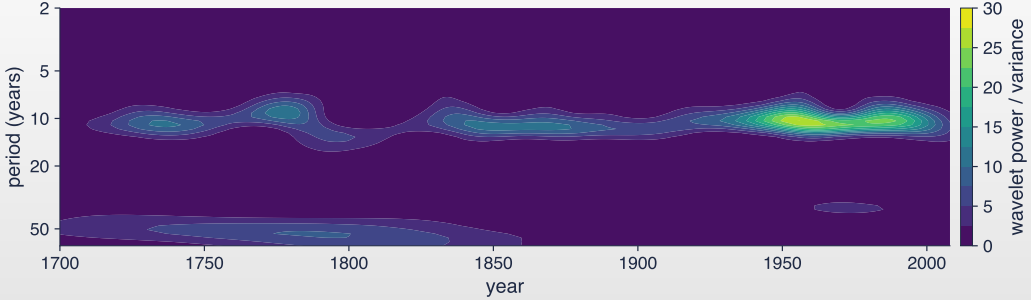
Wavelets (1/2): frequency that changes over time

- The spectrum assumes stationarity: the same cycles at every date
  - the **short-time Fourier transform** (STFT) computes spectra in moving windows of fixed length
- A **wavelet** is a short wave packet; the **continuous wavelet transform** (CWT) stretches it to each period and slides it along the series
  - the result, the **scalogram**, shows the power at each period and each date (Torrence and Compo (1998))
  - short windows for fast cycles, long windows for slow cycles: good resolution at all scales
- Beyond this course; a natural project topic (Python: `pywt`, or the Morlet code of this chapter)

Wavelets (2/2): the continuous wavelet transform

- The **Morlet wavelet**: a complex wave damped by a Gaussian bell
  - $\psi(u) = \pi^{-1/4}\, e^{i\omega_0 u}\, e^{-u^2/2}$, with $\omega_0 = 6$ (Torrence and Compo (1998))
  - u: time measured in units of the scale; ω_0: the number of oscillations inside the bell
- The **transform** at scale s and date τ
  - $W(s,\tau) = \sum_t x_t \, \frac{1}{\sqrt{s}}\, \psi^*\left(\frac{t-\tau}{s}\right)$: a correlation of the series with the wavelet stretched to scale s and centred at τ
  - ψ^*: the complex conjugate of ψ; for $\omega_0 = 6$ the period of the cycle is about $1.03\,s$
- The **scalogram**: the wavelet power $|W(s,\tau)|^2$
  - the analogue of the periodogram, but for each date τ separately

Wavelet power of the sunspots



Morlet wavelet power of the standardised yearly sunspot numbers, 1700–2008, periods 2 to 64 years; bright colours mean high power

TSA_ch12_wavelet

Interpreting the scalogram

- The 11-year band is present throughout, but its strength changes
  - average power in the 8–14-year band: 2.0 in 1795–1830 (the weak Dalton minimum) against 5.7 over the whole sample; the maximum is around 1956
- The periodogram averaged these changes into one peak; the scalogram shows when the cycle was strong or weak
- Edges of the chart are less reliable (the wavelet runs out of data): this region is called the cone of influence

Recap: Long memory, filters, two series, wavelets

- Long memory is a pole at $\nu = 0$: slope $-2d$ on a log–log plot (GPH)
- A filter multiplies the spectrum by its squared gain: differencing and the HP filter reshape cycles
- Coherence is a correlation by frequency; the phase gives the lead or the lag
- Wavelets show how cycles change over time

Possible contribution of AI

- **Code**
  - a first draft of a periodogram, a Daniell smoother, a Welch estimate or a coherence plot
- **Explanation**
  - a second explanation of leakage, aliasing or the bias–variance trade-off of the bandwidth
- **Exploration**
  - spectra of many series at once (load of several countries, GDP of all EU members) and a first reading of the peaks
- Example prompt
  - `Write Python code that reads hourly electricity load, removes the mean, computes the periodogram with a Hann taper and a Welch estimate with 8-week segments, plots both on a log-log scale against the period in hours, and marks 24, 12 and 168 hours.`

Checks you must run

- The units: frequency in cycles per observation or radians; `scipy.signal.welch` returns a one-sided density (twice our f)
- The period: $1/\nu$ in observations; convert to years, quarters or hours yourself
- Parseval: the ordinates must add up to T times the variance; test the code on a cosine with a known frequency
- A peak is a cycle only with a test or a confidence band; a raw periodogram always has tall spikes
- Trends, breaks and filters create low-frequency power and false peaks: check the data and the transformation first
- Every cited reference: it must exist; check the DOI

Key takeaways

- The spectrum splits the variance of a stationary series by frequency; it carries the same information as the ACF
- Read its shape: flat (white noise), falling (persistence), peaked (cycles), a pole at zero (long memory)
- The raw periodogram is unbiased but inconsistent: taper, smooth and add confidence bands
- Real cycles: 11 years in sunspots, about 6.1 years in the US HP cycle of GDP, 24 hours and 1 week in electricity load
- Filters reshape spectra; coherence and phase relate the cycles of two series

Key formulas

Quantity	Formula
Spectral density	$f(\nu) = \sum_h \gamma(h) e^{-2\pi i \nu h}, \quad \gamma(0) = \int_{-1/2}^{1/2} f(\nu)\, d\nu$
AR(1), MA(1)	$\sigma^2/(1 - 2\phi\cos 2\pi\nu + \phi^2), \quad \sigma^2(1 + 2\theta\cos 2\pi\nu + \theta^2)$
ARMA	$\sigma^2 \lvert\theta(e^{-2\pi i\nu})\rvert^2 / \lvert\phi(e^{-2\pi i\nu})\rvert^2$
Periodogram	$I(\nu_j) = \lvert T^{-1/2} \sum_t x_t e^{-2\pi i \nu_j t}\rvert^2, \quad \nu_j = j/T, \quad 2I/f \approx \chi^2_2$
Daniell	$\hat{f}(\nu_j) = \frac{1}{L} \sum_{\lvert k\rvert \le m} I(\nu_{j+k}), \quad L = 2m+1, \quad B = L/T$
95% band	$[2L\hat{f}/\chi^2_{2L}(0.975),\ 2L\hat{f}/\chi^2_{2L}(0.025)]$
Linear filter	$f_y(\nu) = \lvert A(\nu)\rvert^2 f_x(\nu), \quad \lvert 1 - e^{-2\pi i\nu}\rvert^2 = 4\sin^2(\pi\nu)$
Coherence, phase	$\rho^2_{xy} = \lvert f_{xy}\rvert^2/(f_x f_y), \quad \varphi = \arg f_{xy}, \quad \text{lag} = \lvert\varphi\rvert/(2\pi\nu)$

Self-assessment (1/2)

- **Question**: weekly data show a peak at $\nu = 0.0192$. What is the period?
  - **Answer**: $1/0.0192 \approx 52$ weeks: an annual cycle
- **Question**: what are $f(0)$ and $f(1/2)$ of an AR(1) with $\phi = -0.5$, $\sigma^2 = 1$?
  - **Answer**: $f(0) = 1/1.5^2 = 0.44$ and $f(1/2) = 1/0.5^2 = 4$: power at high frequencies
- **Question**: why does a longer sample not make the raw periodogram more precise?
  - **Answer**: each ordinate still has variance about f^2; more data only adds ordinates; precision needs averaging

Self-assessment (2/2)

- **Question**: a monthly series observed only every second month has a 3-month cycle. Where does it appear?
  - **Answer**: $\nu = 2/3$ cycles per two months is above $1/2$; it folds to $1/3$: a false cycle of 3 observations, i.e. 6 months
- **Question**: what does the first difference do to a cycle of 40 quarters?
  - **Answer**: it multiplies its power by $4\sin^2(\pi/40) = 0.025$: the cycle almost disappears
- **Question**: the log–log periodogram of a series has slope -0.6 near zero. What is d?
  - **Answer**: slope $= -2d$, so $d = 0.3$: stationary long memory (Chapter 8)
- **Question**: the squared coherence of two series is 0.9 at 5-year cycles and 0.1 at monthly cycles. What does it mean?
  - **Answer**: their business cycles move together; their short-run noise does not
- Next: Chapter 13, speculative bubbles and LPPL models

References (1/2)

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